Big Data Analytics with MongoDB

MongoDB

- MongoDB is a document-oriented NoSQL database used for high volume data storage. It is a cross-platform document database that provide high performance, high availability and easy scalability.
- It works on the concepts of collections and documents. Collection means a group of MongoDB documents.
- Collection do not enforce schema. Document within a collection can have different fields
- Document is a set of key-value pairs. Documents have a dynamic schema.

Comparison between RDMS & MongoDB

RDBMS	MongoDB
Database	Database
Table	Collection
Tuple/Row	Document
column	Field
Table Join	Embedded Documents
Primary Key	Primary Key (Default key _id provided by mongodb itself)
Database Server and Client	
Mysqld/Oracle	mongod
Mysqld	mongo

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Features of MongoDB

- It is schemaless document database
- It has clear object structure
- It has no complex join
- It has deep-query ability which supports document-based language
- It is more scalable
- It uses internal memory for storing working data sets.

Why MongoDB?

- It is document oriented storage, it sores data in the form of JSON – JavaScript Object Notation
- It provides index to any attributes
- It provides replication and high availability of data
- It has auto-sharding mechanism
- It has powerful queries

Sample Document

```
{
title: 'Titanic',
year: 1997,
genres: [ 'Drama', 'Romance' ],
rated: 'PG-13',
languages: [ 'English', 'French', 'German', 'Swedish', 'Italian', 'Russian' ],
released: ISODate("1997-12-19T00:00:00.000Z"),
awards: {
wins: 127,
nominations: 63,
text: 'Won 11 Oscars. Another 116 wins & 63 nominations.'
},
cast: [ 'Leonardo DiCaprio', 'Kate Winslet', 'Billy Zane', 'Kathy Bates' ],
directors: [ 'James Cameron' ]
}
```

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MongoDB Installation

- To install MongoDB on Windows, first download the latest release of MongoDB from http://www.mongodb.org/downloads.
- Once you have finished installed the mongodb, to get your Windows version, open command prompt and execute the following command. “mongod –version”

MongoDB Shell

- The MongoDB shell is an interactive JavaScript-based interface to interact with a MongoDB database.
- It allows users to execute database commands, perform CRUD (Create, Read, Update, Delete) operations, and manage the database server.
- Thus, to work with MongoDB smoothly you need to install MongoDB Shell.
- You can download the MongoDB Shell via this link: https://www.mongodb.com/try/download/shell Note: make sure you choose msi package type to get the executable file setup.
- Once you have finished the installation, to get access into MongoDB via mongodb sell, go to **cmd** commond prompt, type the command “mongosh” then press enter.

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MongoDB - Creating new database

The use Command

MongoDB use DATABASE_NAME to create database. The command will create a new database if it doesn't exist, otherwise it will return the existing database.

Syntax

Basic syntax of ***use DATABASE*** statement is as follows:

Use database FDS

To check your currently selected database, use the command db

→ db

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MongoDB - Commands

The show Command

- To see a list of database, use the command **show dbs**.

→Show dbs

The insert Command

- NOTE: A blank created database is not listed in the available database list, you need to insert at least one document into it.
- `>db.studentList.insert({"name":"Ali Khelef",”age”:39})`

Drop Database Command

MongoDB use db.dropDatabase() command to drop an existing database.

Syntax

Basic syntax of dropDatabase() command is as follows:

`>db.dropDatabase()`

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MongoDB - Commands

Create collection Command

The createCollection() Method

MongoDB use db.createCollection(name, options) to create collection.

Syntax

Basic syntax of createCollection() command is as follows:

>db.createCollection(“Staff”)

NOTE: In MongoDB, you don't need to create collection. MongoDB creates collection automatically, when you insert some document.

Drop Collection Command

The drop() Method

MongoDB use db.collection_name.drop() command to drop an existing collection.

Syntax

Basic syntax of drop() command is as follows:

>db.Staff.drop()

MongoDB - CRUD Commands

CRUD operations

- Create
  - db.collection.insert(<document>)
  - db.collection.save(<document>)
  - db.collection.update(<query>, <update>, { upsert: true })
- Read
  - db.collection.find(<query>, <projection>)
  - db.collection.findOne(<query>, <projection>)
- Update
  - db.collection.update(<query>, <update>, <options>)
- Delete
  - db.collection.remove(<query>, <justOne>)

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MongoDB - CRUD Commands

Select All Documents

To select all documents in the collection, pass an empty document as the query filter document to the db.collection.find() method

That is: db.collection_name.find({}}

For example: db.students.find({})

```
HR> db.students.find({})
[
  {
    _id: ObjectId("63a05be2d763d14f15fa69cd"),
    name: 'Ali',
    age: 35,
    sex: 'Male'
  },
  {
    _id: ObjectId("63a05be2d763d14f15fa69ce"),
    name: 'Asha',
    age: 40,
    sex: 'Female'
  }
]
```

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MongoDB - QUERY Commands

Query Operators

Name	Description
$eq	Matches value that are equal to a specified value
$gt, $gte	Matches values that are greater than (or equal to a specified value
$lt, $lte	Matches values less than or (equal to) a specified value
$ne	Matches values that are not equal to a specified value
$in	Matches any of the values specified in an array
$nin	Matches none of the values specified in an array
$or	Joins query clauses with a logical OR returns all
$and	Join query clauses with a loginal AND
$not	Inverts the effect of a query expression
$nor	Join query clauses with a logical NOR
$exists	Matches documents that have a specified field

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MongoDB - CRUD Commands

Fitering Datat based on logical condition

For example to select all students from students collection whose age is greater than 36 years.

db.students.find({“age”:{“$gt”:36}})

```
HR> db.students.find({"age":{"$gt":36}})
[
  {
    _id: ObjectId("63a05be2d763d14f15fa69ce"),
    name: 'Asha',
    age: 40,
    sex: 'Female'
  }
]
```

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NECTA-TASK: Group Exercise

(5 students @ 30 marks for one week, deadline: 6th Jan, 2025 at 16:00hrs)

Instruction for Submission: Prepare a short video clip demonstrating the way you achieve your answer. Then, submit a paper group list with your video clip before 6th Jan, 2025 at 16:00hrs

https://onlinesys.necta.go.tz/results/2022/acsee/index.htm

Problem: The director of NECTA want to get information about the performance of each subject in the ACESS school all over the country.

1. Assume you have been asked to extract a school result html page and load it into mongoDB. How can you do this task? (Hint: use any software tool of your choice that can help you to achieve your answer)